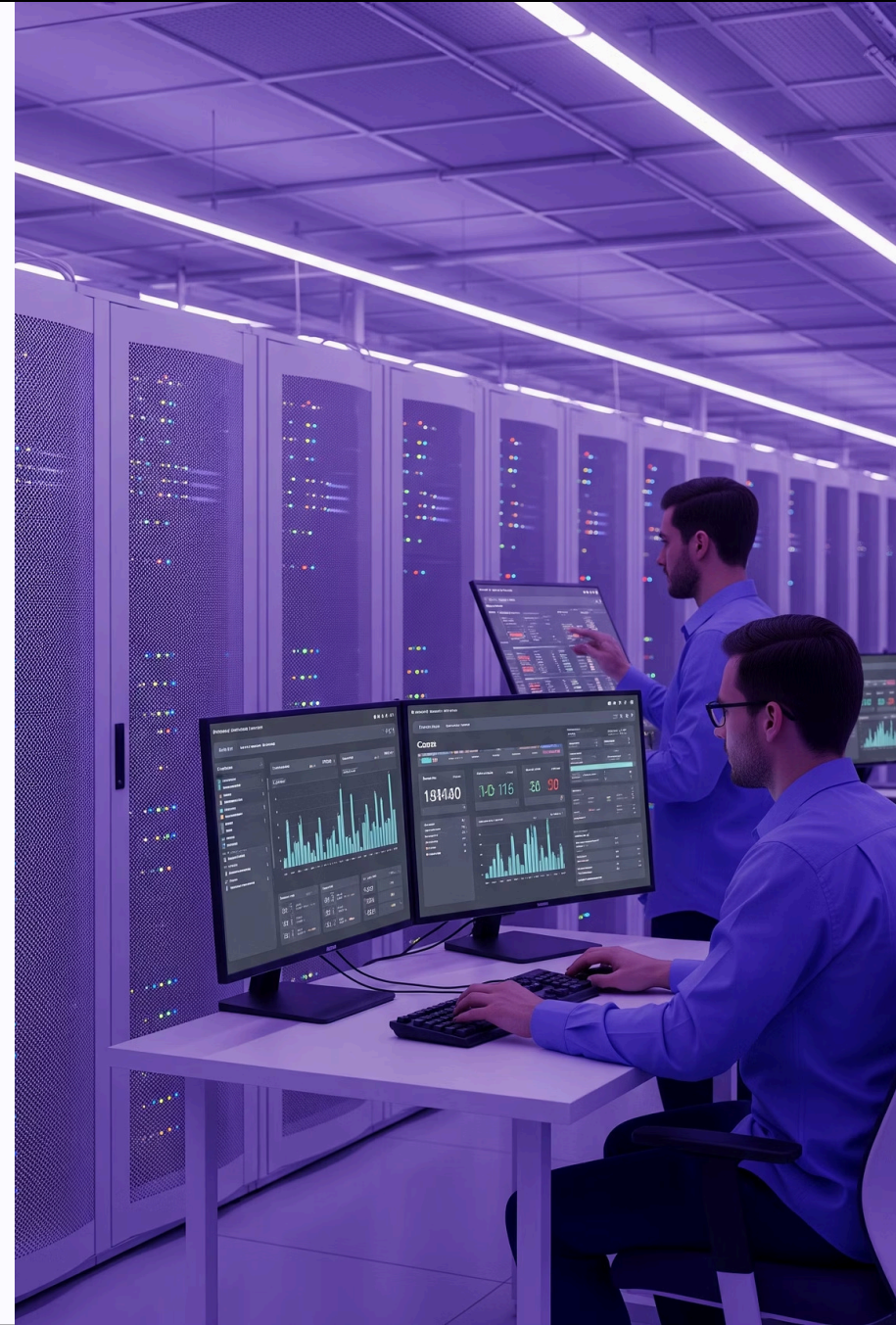


Review of Cost Optimization in AI Infrastructure

Maximizing performance per dollar for compute, storage, networking, and serving infrastructure at scale



Why Costs Skyrocket in AI

Training

Thousands of GPUs running continuously for weeks at a time

Inference

Billions of daily queries requiring specialized hardware

Storage

Petabytes of raw and processed data requiring high-performance storage

Networking

High interconnect demand requiring specialized solutions like InfiniBand and NVLink

The scale of modern AI operations creates exponential cost challenges across all infrastructure layers

Compute Cost Strategies

Spot / Preemptible Instances

Save 60-80% on training costs using interruptible compute with checkpointing

Right-sizing

Match model size and hardware to workload requirements

Auto-scaling Clusters

Scale inference resources based on demand patterns

Model Optimization

Distillation and quantization to reduce compute requirements



Storage Cost Strategies



Implement Tiered Storage

Hot data on NVMe, warm data on object storage, cold data in archives



Deduplication + Compression

Reduce storage footprint of training datasets and checkpoints



Delete Stale Data

Implement automated lifecycle policies for checkpoints and logs



Object vs. Block Storage

Use S3/GCS instead of expensive block storage when possible

Networking Cost Strategies

Co-locate compute and data

Position compute resources close to data to minimize costly data transfer fees

Use private interconnects

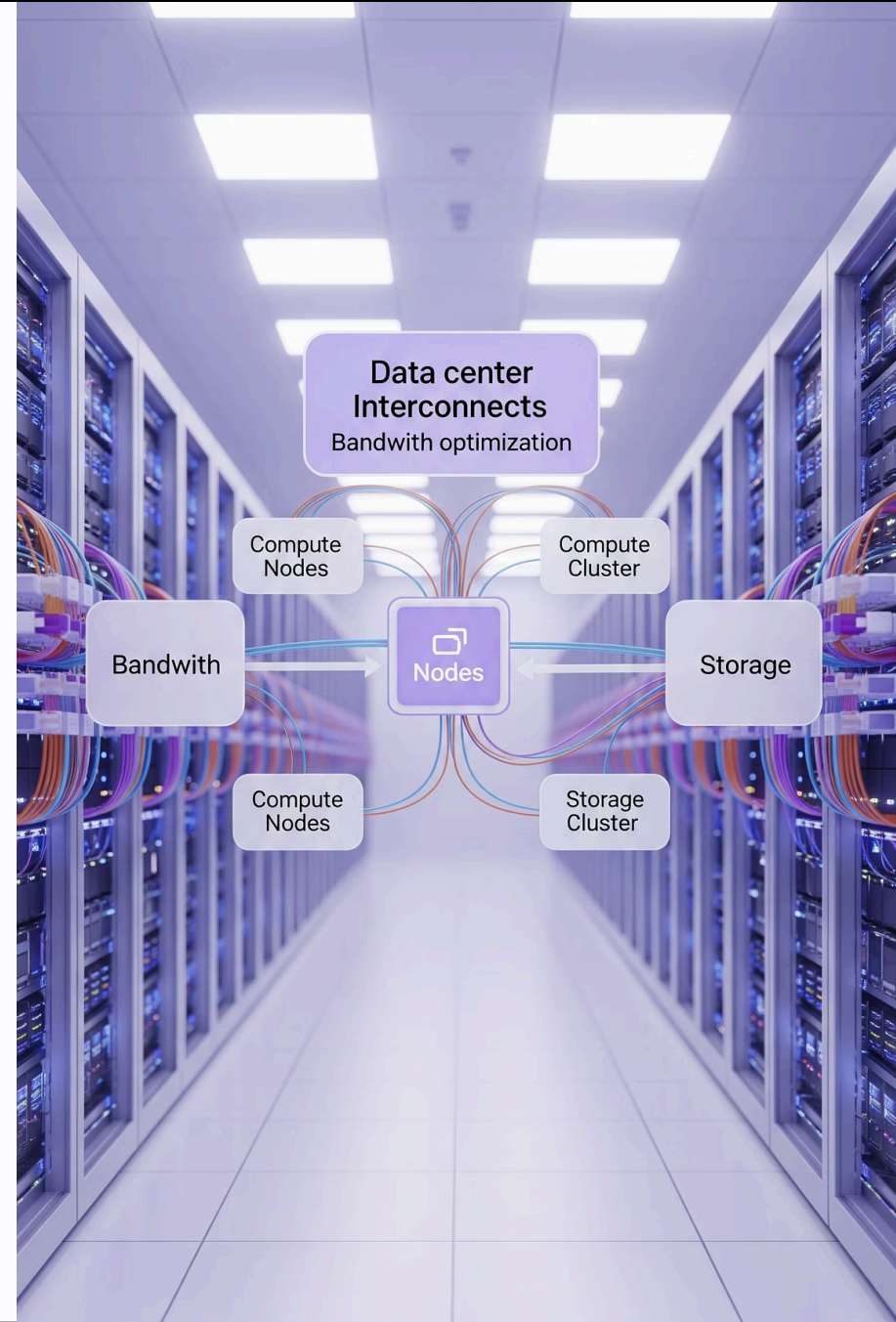
Direct connections between data centers cost less than public internet egress

Optimize bandwidth usage

Implement gradient compression techniques in distributed training

Leverage CDN caching

Cache frequent inference API responses at edge locations to reduce backbone traffic



Inference Cost Optimization



Model Tiering

Use smaller models for common queries, routing to larger models only for complex edge cases



Response Caching

Cache frequent embedding search results and RAG responses to avoid recomputation



Hardware Optimization

Deploy TensorRT, ONNX Runtime, or other hardware-specific optimizations



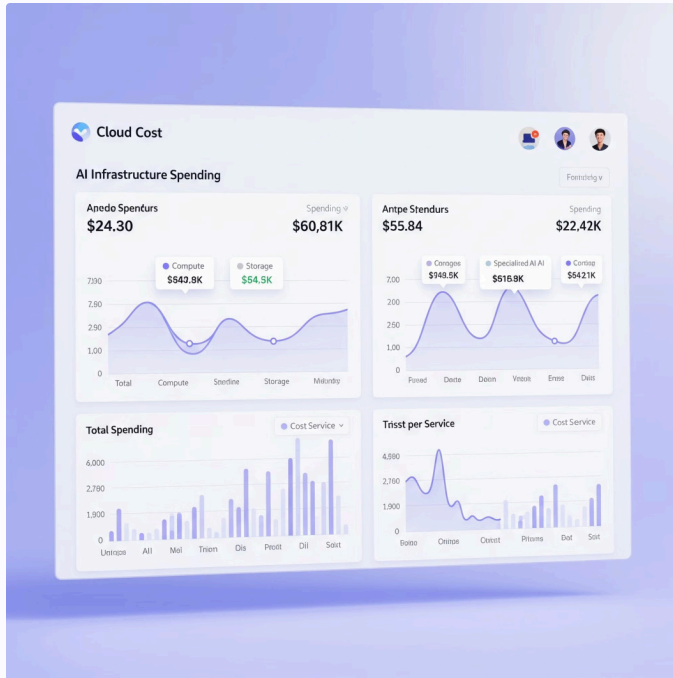
Batching

Group inference requests where latency requirements permit

Cloud Billing & FinOps

FinOps = Cloud Financial Management

A discipline bringing financial accountability to variable cloud spend through cross-functional collaboration



Essential FinOps Tools

- AWS Cost Explorer, GCP Billing, Azure Cost Management
- Budgets + alerts to detect runaway training jobs
- Tagging resources for attribution
- Chargeback/showback mechanisms for multi-team GPU clusters
- Cost anomaly detection

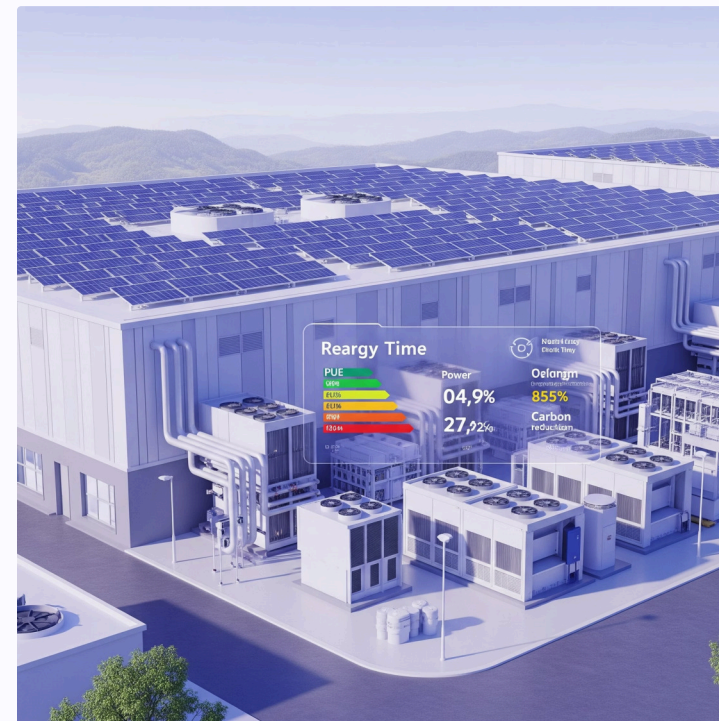
Sustainability & Cost: The Green Link

Energy drives both cost and carbon footprint

Efficient models (FP8, INT4) reduce both costs and emissions

Green AI provides both financial and regulatory advantages

DeepMind cut Google data center cooling costs by 40% using RL



i When optimizing for cost, you're often simultaneously optimizing for sustainability—making the business case stronger.

Case Study: Netflix AI Infrastructure



The Challenge

Scaling personalized recommendations to 200M+ users while controlling costs

The Approach

Two-tiered recommendation system: low-cost candidate generation + expensive ranking

The Result

Hybrid model significantly cuts GPU costs at scale while maintaining quality

Key Learning: Extensive A/B testing framework prevents costly wasted experiments by validating cost-benefit before full deployment

Case Study: Tesla FSD Infrastructure

The Challenge

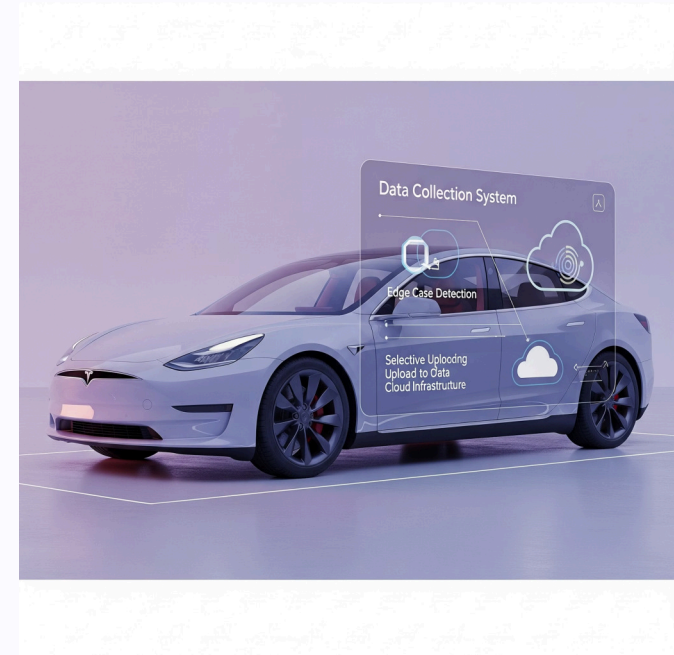
Managing massive fleet data without breaking the storage bank

The Approach

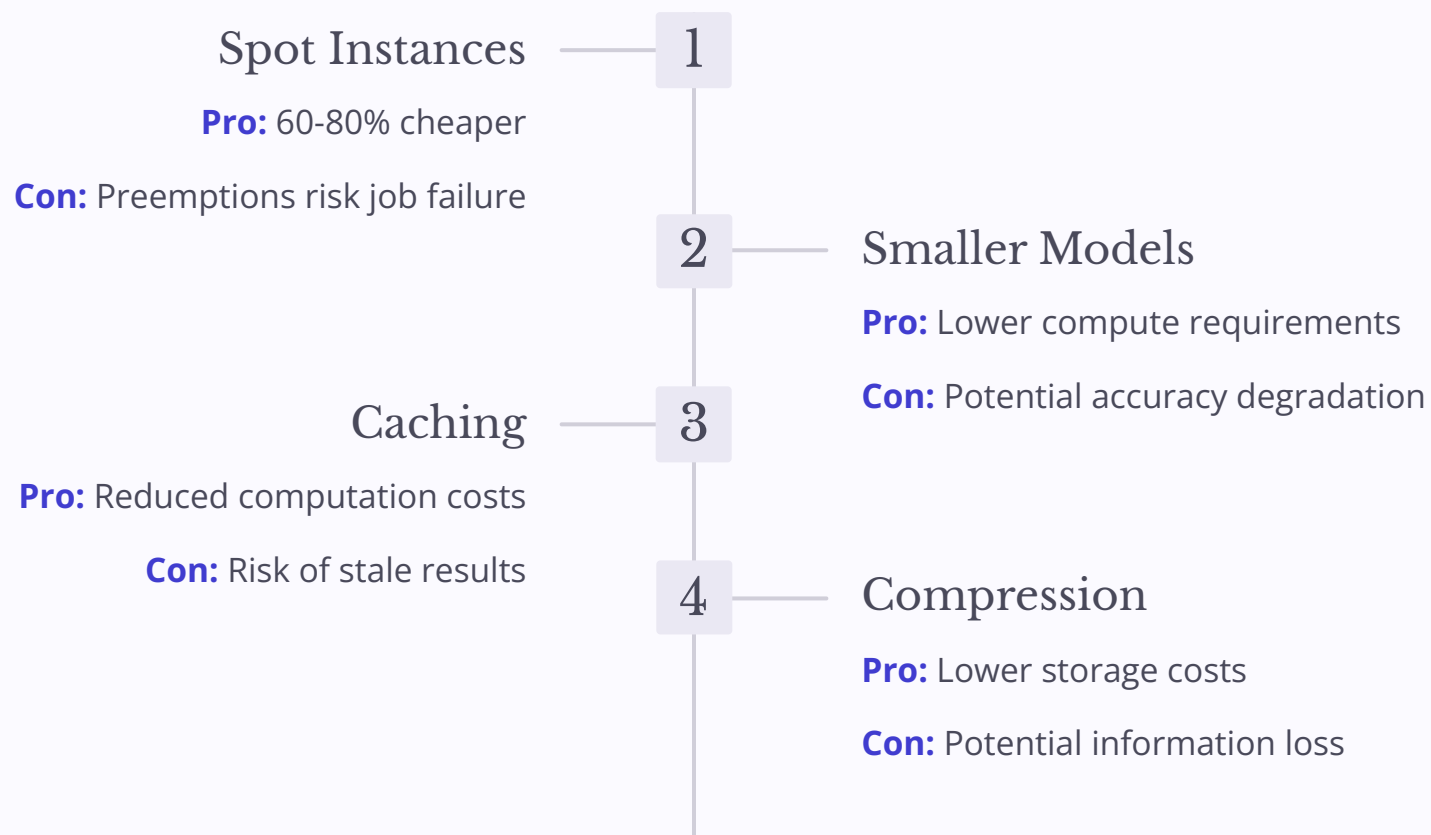
- Selective logging: only edge cases uploaded to cloud
- Smart filtering at vehicle level
- Custom Dojo silicon designed specifically to reduce training costs

The Result

Saved petabytes of unnecessary storage while still capturing critical training data



Trade-Offs in Cost Optimization



The balancing act: cost vs. accuracy vs. reliability

Every Infra Layer Has Cost Levers to Pull



Cost optimization requires a holistic approach across all infrastructure components. Each layer presents unique opportunities for efficiency gains.

Key Lessons in AI Infrastructure Cost Optimization

1

Beyond Hardware

Cost optimization isn't just about cheaper hardware—it's about **strategic design choices** in your architecture

2

Model Optimization

Distillation, quantization, and pruning typically yield the **biggest savings** compared to other approaches

3

FinOps Discipline

Rigorous monitoring and financial governance **prevent budget overruns** before they happen

4

Dual Benefits

Cost efficiency and sustainability increasingly **overlap in business value**, creating win-win opportunities

Summary: Building Cost-Aware AI Infrastructure

AI Infrastructure Cost Drivers

- Compute: GPUs, CPUs, specialized accelerators
- Storage: Training data, checkpoints, artifacts
- Networking: Data transfer, interconnects
- Inference: Serving infrastructure

Key Optimization Strategies

- Spot instances, tiered storage, and effective caching
- Model compression, distillation, and quantization
- FinOps practices and continuous monitoring

Capstone Success Metric:

Building **cost-aware infrastructure designs** that balance performance needs with financial constraints



Discussion: Strategic Cost Trade-offs

"If you were CTO of a startup, would you cut costs by using smaller models (lower accuracy) or spot instances (less reliable compute) — and why?"

Smaller Models

- Predictable performance
- Faster inference
- Lower memory requirements
- But: potentially lower accuracy

Spot Instances

- Same model quality
- Significant cost savings (60-80%)
- Checkpointing mitigates risks
- But: potential training interruptions

Consider how your specific use case, risk tolerance, and performance requirements shape this decision